The documentation and process conversion measures necessary to comply with this revision shall be completed by 6 June 2006.

INCH-POUND
MIL-PRF-19500/444J
6 March 2006
SUPERSEDING
MIL-PRF-19500/444H
1 June 2004

PERFORMANCE SPECIFICATION SHEET

* SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING, SCHOTTKY, TYPES 1N5711-1, 1N5711UR-1, 1N5711UB, 1N5711UBCA, 1N5711UBD, 1N5711UBCC, 1N5712-1, 1N5712UR-1, 1N5712UB, 1N5712UBCA, 1N5712UBD, 1N5712UBCC, 1N6857-1, 1N6857UR-1, 1N6858-1, AND 1N6858UR-1, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for Schottky barrier diodes. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1, similar to DO-35, figure 2, DO-213AA, (UR) surface mount package, figure 3 (UB) surface mount, and figure 4, JANHC and JANKC.

1.3 Maximum ratings.

| Type | $V_{RWM}$ | $I_{O1}$ | $I_{O2}$ (1) | $T_{STG}$ and $T_J$ | $R_{\Theta JL}$ L = .375 inch (9.52 mm) (2) | $R_{\Theta JEC}$ (UR) | $R_{\Theta JSP}$ (UBx) | ESDS class |
|---|---|---|---|---|---|---|---|---|
| | V(pk) | mA dc | mA dc | | (leaded) | (UR) | (UBx) | |
| 1N5711-1 | 50 | 33 (3) | 5 | -65°C to +150°C | 250 | | | 1 |
| 1N5711UR-1 | 50 | 33 (4) | 5 | | | 100 | | 1 |
| 1N5711UB | 50 | 33 (4) | 5 | | | | 100 | 1 |
| 1N5711UBCA | 50 | 33 (4) | 5 | | | | 100 | 1 |
| 1N5711UBCC | 50 | 33 (4) | 5 | | | | 100 | 1 |
| 1N5711UBD | 50 | 33 (4) | 5 | | | | 100 | 1 |
| 1N6858-1 | 50 | 75 (5) | 10 | | 250 | | | 2 |
| 1N6858UR-1 | 50 | 75 (6) | 10 | | | 100 | | 2 |
| 1N5712-1 | 16 | 75 (5) | 20 | | 250 | | | 1 |
| 1N5712UR-1 | 16 | 75 (6) | 20 | | | 100 | | 1 |
| 1N5712UB | 16 | 75 (6) | 20 | | | | 100 | 1 |
| 1N5712UBCA | 16 | 75 (6) | 20 | | | | 100 | 1 |
| 1N5712UBCC | 16 | 75 (6) | 20 | | | | 100 | 1 |
| 1N5712UBD | 16 | 75 (6) | 20 | | | | 100 | 1 |
| 1N6857-1 | 16 | 150 (7) | 50 | | 250 | | | 2 |
| 1N6857UR-1 | 16 | 150 (8) | 50 | | | 100 | | 2 |

See notes on next page.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.



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1.3 Maximum ratings – Continued.

(1) Maximum $I_O$ rating to ensure $\tau_{CL}$ compliance (< 100 ps).
(2) Lead length = .375 inch (9.52 mm).
(3) At $T_L$ = +130°C and L = .375 inch, derate $I_O$ to 0 at +150°C.
(4) At $T_{EC}$ and $T_{SP}$ = +140°C, derate $I_O$ to 0 at +150°C.
(5) At $T_L$ = +110°C and L = .375 inch, derate $I_O$ to 0 at +150°C.
(6) At $T_{EC}$ and $T_{SP}$ = +130°C, derate $I_O$ to 0 at +150°C.
(7) At $T_L$ = +70°C and L = .375 inch, derate $I_O$ to 0 at +150°C.
(8) At $T_{EC}$ and $T_{SP}$ = +110°C, derate $I_O$ to 0 at +150°C.

1.4 Primary electrical characteristics.

| Type (1) | $V_{(BR)1}$ $I_R$ = 10 µA dc (min) | $V_{F1}$ $I_F$ = 1 mA dc (max) | $V_{F2}$ $I_F$ = 15 mA dc (max) | $I_{R1}$ $V_R$ = 50 V dc (max) | C $V_R$ = 0, f = 1 MHz $V_{sig}$ = 50 m V(pk) (max) | $\tau_{CL}$ (max) | | $Z_{\Theta JX}$ (max) |
|---|---|---|---|---|---|---|---|---|
| | V dc | V dc | V dc | nA dc | pF | ps | mA | °C/W |
| 1N5711-1 | 70 | 0.410 | 1.0 | 200 | 2.0 | 100 | 5 | 40 |
| 1N6858-1 | 70 | 0.360 | 0.65 | 200 | 4.5 | 100 | 10 | 40 |

| Type (1) | $V_{(BR)1}$ $I_R$ = 10 µA dc (min) | $V_{F1}$ $I_F$ = 1 mA dc (max) | $V_{F2}$ $I_F$ = 35 mA dc (max) | $I_{R1}$ $V_R$ = 16 V dc (max) | C $V_R$ = 0, f = 1 MHz $V_{sig}$ = 50 mv(pk) (max) | $\tau_{CL}$ (max) | | $Z_{\Theta JX}$ (max) |
|---|---|---|---|---|---|---|---|---|
| | V dc | V dc | V dc | nA dc | pF | ps | mA | °C/W |
| 1N5712-1 | 20 | 0.410 | 1.0 | 150 | 2.0 | 100 | 20 | 40 |
| 1N6857-1 | 20 | 0.350 | 0.75 | 150 | 4.5 | 100 | 50 | 40 |

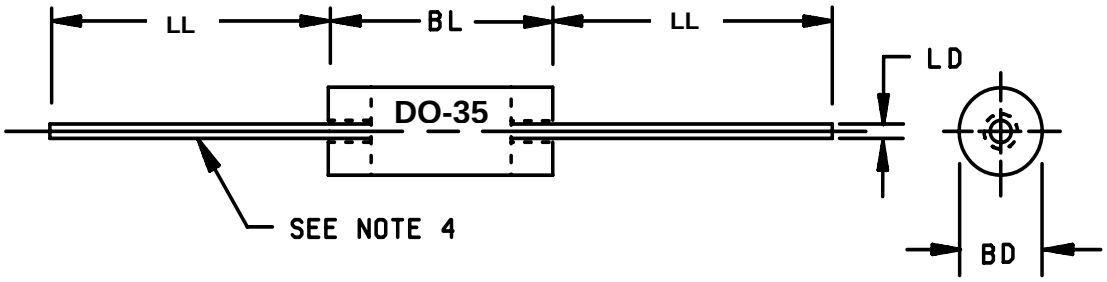
(1) Electrical characteristics are for all package styles.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| Ltr | Min | Max | Min | Max | Notes |
| BD | .068 | .076 | 1.73 | 1.93 | 2, 3 |
| BL | .150 | .170 | 3.81 | 4.32 | 2 |
| LD | .014 | .022 | 0.36 | 0.56 | |
| LL | 1.000 | 1.500 | 25.40 | 38.10 | |

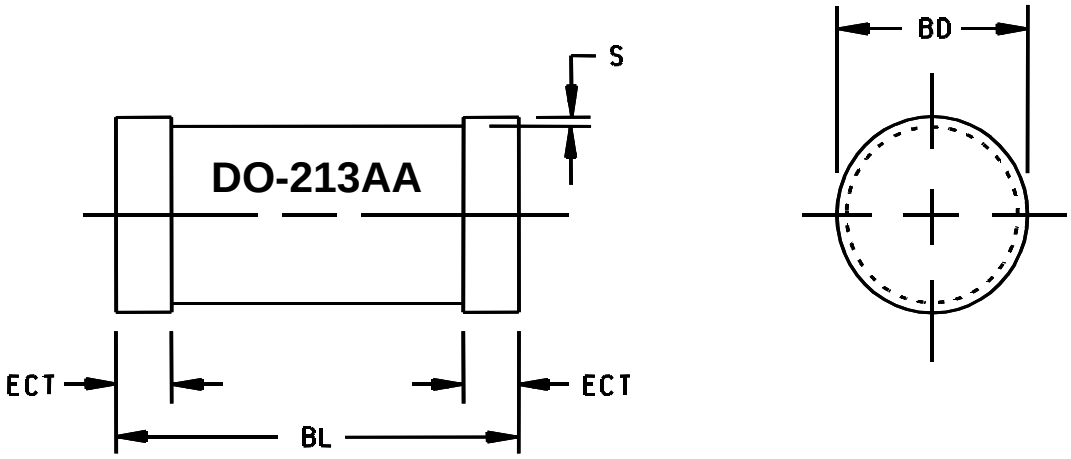
NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions BL and LD include all components of the diode periphery except the sections of the leads over which the diameter is controlled.
3. Dimension BD shall be measured at the largest diameter.
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

1N5711-1, 1N5712-1, 1N6857-1, and 1N6858-1

FIGURE 1. Physical dimensions, (similar to DO-35).



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| BD | .063 | .067 | 1.60 | 1.70 | |
| BL | .130 | .146 | 3.30 | 3.71 | |
| ECT | .016 | .022 | 0.41 | 0.55 | |
| S | .001 Min | | 0.03 Min | | |

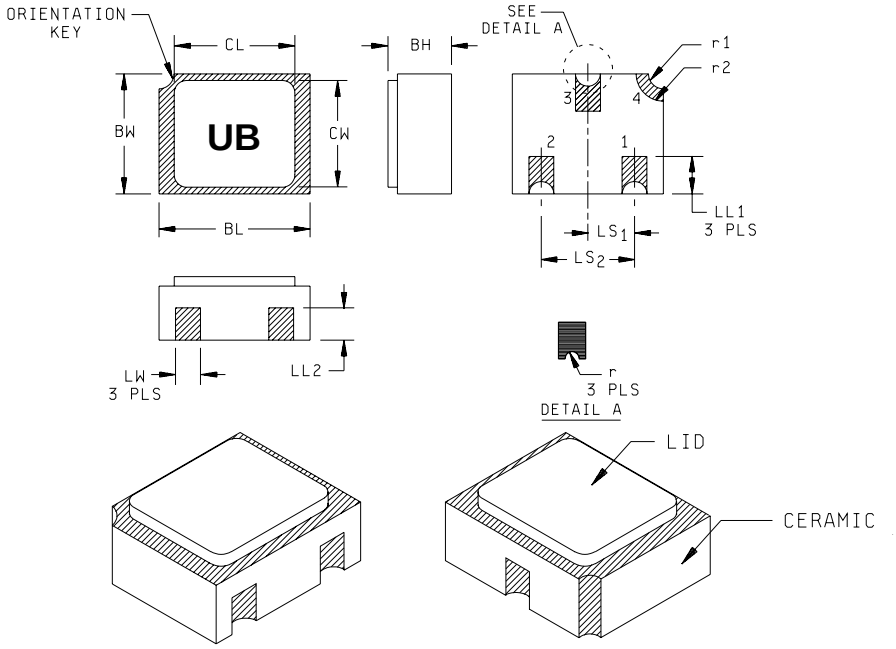
NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

1N5711UR-1, 1N5712UR-1, 1N6857UR-1, and 1N6858UR-1

FIGURE 2. Physical dimensions, (DO-213AA).

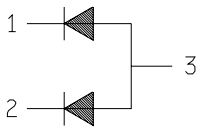


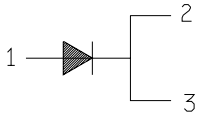
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| Symbol | Dimensions | | | | Symbol | Dimensions | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| | Min | Max | Min | Max | | Min | Max | Min | Max |
| BH | .046 | .056 | 1.17 | 1.42 | $LS_1$ | .035 | .040 | 0.89 | 1.02 |
| BL | .115 | .128 | 2.92 | 3.25 | $LS_2$ | .071 | .079 | 1.81 | 2.01 |
| BW | .085 | .108 | 2.16 | 2.74 | LW | .016 | .024 | 0.41 | 0.61 |
| CL | | .128 | | 3.25 | r | | .008 | | 0.20 |
| CW | | .108 | | 2.74 | r1 | | .012 | | 0.31 |
| LL1 | .022 | .038 | 0.56 | 0.96 | r2 | | .022 | | 0.56 |
| LL2 | .017 | .035 | 0.43 | 0.89 | | | | | |

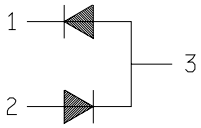
1N5711UBCA
1N5712UBCA

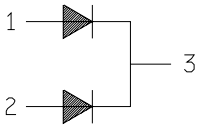



1N5711UB
1N5712UB

1N5711UBD
1N5712UBD



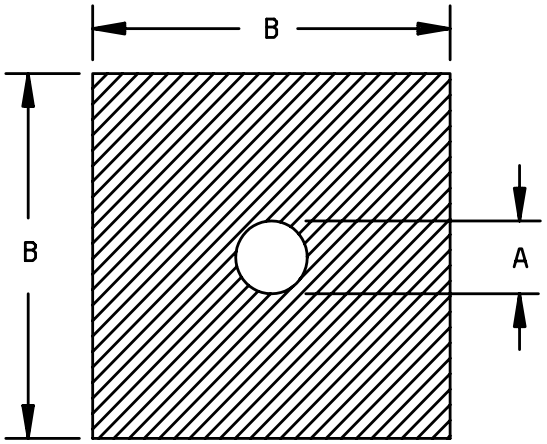

1N5711UBCC
1N5712UBCC

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 3. Physical dimensions, surface mount (UB version).



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| 1N5711 Dimensions | | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| Ltr | Min | Max | Min | Max | Notes |
| A | .004 | .005 | 0.102 | 0.127 | |
| B | .0130 | .0170 | 0.330 | 0.432 | |

| 1N5712 Dimensions | | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| Ltr | Min | Max | Min | Max | Notes |
| A | .0029 | .0035 | 0.074 | 0.089 | |
| B | .0130 | .0170 | 0.330 | 0.432 | |

| 1N6857, 1N6858 Dimensions | | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| Ltr | Min | Max | Min | Max | Notes |
| A | .0055 | .0065 | 0.140 | 0.165 | |
| B | .0130 | .0170 | 0.330 | 0.432 | |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Element evaluation accomplished utilizing TO-5 package.
3. The physical characteristics of the die are:
   Metallization:
   Top (anode): Al.
   Back (cathode): Au.
   Al thickness: 25,000 Å minimum.
   Gold thickness: 4,000 Å minimum.
   Chip thickness: .010 inch (0.25 mm) ±.002 inch (0.05 mm).
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWING

DEFENSE ELECTRONICS SUPPLY CENTER (DESC)

C68001 - Test Fixture for Effective Minority Carrier Lifetime.

(Copies of DESC Drawing C68001 are available from Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail Semiconductor@dscc.dla.mil.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

| | |
|---|---|
| EC | End-cap. |
| $\tau_{CL}$ | Effective carrier lifetime. |
| UR | Surface mount case outline, round end-cap. |

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1, 2, 3, and 4 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).



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3.4.2 Diode construction. All devices (except UB version) shall be metallurgically bonded, double plug construction in accordance with the requirements of MIL-PRF-19500. All glass diodes shall be designed with sufficient thermal compensation in the axial direction to optimize tensile and compressive stresses. Dimensional analysis is required of all materials used to achieve axial thermal compensation. Dimensional tolerances and corresponding coefficient of thermal expansion (CTE) shall be documented on the DSCC Design and Construction Form 36D and shall be approved by the qualifying activity to maintain qualification. Dimensional tolerances shall be sufficiently tight enough to prevent excessive stresses due to the inherent CTE mismatch. The UB devices shall be eutectically mounted and wire bonded in a ceramic package.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I.

3.6 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

* 3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL-PRF-19500. The polarity shall be indicated with a contrasting color band to denote the cathode end. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, or JS, respectively. The part number may be reduced to J5711, JX5711, JV5711, or JS5711. No color coding shall be permitted for part numbering.

3.7.1 Polarity. The polarity of axial leaded devices shall be indicated with a contrasting color band to denote the cathode end. No color coding will be permitted for the part number.

* 3.7.1.1 UB devices. 'UB' devices do not require polarity marking.

* 3.7.2 UR devices. For 'UR' version devices only, all marking, except polarity may be omitted from the body, but shall be retained on the initial container. Polarity marking of 'UR' devices shall consist, as a minimum, a band or three contrasting dots spaced equally around the periphery of the cathode.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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***** 4.2.1.1 Group E thermal impedance. Each supplier shall submit a thermal impedance ($Z_{\Theta JX}$) histogram of the entire qualification lot. The histogram data shall be taken prior to the removal of devices that are atypical for thermal impedance. Thermal impedance curves (from $Z_{\Theta JX}$ test pulse time to $R_{\Theta JX}$ minimum steady-state time) of the best device in the qual lot and the worst device in the qual lot (that meets the supplier proposed screening limit), or from the thermal grouping, shall be submitted. The optimal test conditions and proposed initial thermal impedance screening limit shall be provided in the qualification report. Data indicating how the optimal test conditions were derived for $Z_{\Theta JX}$ shall also be submitted. The proposed specification maximum thermal impedance curve shall be submitted. The qualifying activity may approve a different $Z_{\Theta JX}$ limit not to exceed the specification's thermal curve for conformance inspection end-point measurements as applicable. Equivalent data, procedures, or statistical process control plans may be used for part, or all, of the above requirements. The approved thermal impedance conditions and limit for $Z_{\Theta JX}$ shall be used by the supplier in screening and table I, subgroup 2. The approved thermal resistance conditions for $R_{\Theta JX}$ shall be used by the supplier for conformance inspection. For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal impedance curves.

4.2.2 JANHC and JANKC devices. Qualification shall be in accordance with MIL-PRF-19500.

* 4.3 Screening. Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS | JANTXV and JANTX level |
| 1a | Required | Not required |
| 1b | Required | Required (JANTXV only) |
| 2 | Not required | Not required |
| 3a<br>3b<br>(1) 3c | Required<br>Not applicable<br>Required (see 4.3.3) | Required<br>Not applicable<br>Required (see 4.3.3) |
| 4 | Not applicable | Not applicable |
| 5 | Required for UB | Not applicable |
| 6 | Not applicable | Not applicable |
| 7a | Optional | Optional |
| (2) 7b | Optional | Optional |
| 8 | Required | Not required |
| 9 and 10 | Not applicable | Not applicable |
| 11 | Required $I_{R1}$ and $V_{F1}$ | Required $I_{R1}$ and $V_{F1}$ |
| 12 | Required, see 4.3.1 | Required. See 4.3.1, t = 48 hours |
| (3) 13 | Required. Subgroup 2 and 3 of table I herein; $\Delta I_{R1} \leq 100$ percent of initial reading or 25 nA whichever is greater; $\Delta V_{F1} \leq \pm 20$ mV dc. | Subgroup 2 of table I herein; $\Delta I_{R1} \leq 100$ percent of initial reading or 50 nA whichever is greater; $\Delta V_{F1} \leq \pm 40$ mV dc. |
| 14a<br>14b | Required for UB<br>Required (2) | Required for UB<br>Required (2) |
| 15 | Required | Not required |
| 16 | Required | Not required |

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with table IV of MIL-PRF-19500, screen 3 prior to this thermal test.

* (2) For clear glass diodes, test may be performed any time after temperature cycling.

(3) Test within 24 hours after removal from test.



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4.3.1 Burn-in conditions. Burn-in conditions are as follows: High temperature reverse bias (HTRB), method 1038 of MIL-STD-750, test condition A, $V_R$ = 40 V; $T_A$ = +150°C (min) for 1N5711 and 1N6858. $V_R$ = 12.8 V, $T_A$ = +150°C (min) for 1N5712 and 1N6857.

4.3.2 Screening (JANHC or JANKC). Screening of JANC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

* 4.3.3 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, as applicable, using the guidelines in that method for determining $I_H$ and $I_M$. $t_{MD}$ shall be 70 μs maximum, $t_H$ shall be 10 ms maximum. See group E, subgroup 4 of table II herein.

* 4.3.4 JAN testing. JAN level product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection . Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (for JANS) and table VIb (for JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Required for UB. |
| B3 | See 4.5.5 | Preconditioning and deltas for temperature cycling. |
| B4 | 1037 | $I_{O1}$ at rated $V_{RWM}$; f = 60 Hz; $T_A$ = room ambient as defined in 4.5 of MIL-STD-750; $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2,000 cycles. |
| B5 | 1038 | HTRB in accordance with 4.3.1. |
| B6 | 3101 or 4081 | $R_{\Theta JL}$ = 250°C/W .375 inch (9.52 mm) lead length (non-surface mount). $R_{\Theta EC}$, $R_{\Theta JSP}$ = 100°C/W (UR suffix devices). |



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4.4.2.2 Group B inspection, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | See 4.5.5 | Preconditioning and deltas for temperature cycling. |
| B2 | 1056 | 0°C to +100°C, 10 cycles, n = 22, c = 0. |
| B2 | 1051 | -55°C to +150°C, 25 cycles, n = 22, c = 0. |
| B3 | 1038 | HTRB, in accordance with 4.3.1. |
| B4 | 2101 | See 4.5.4. (Not applicable to UB.) |
| B4 | 2075 | See 4.5.4.1. (Scribe and break not applicable to UB.) |
| B6 | 1032 | $T_A$ = +150°C. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | See 4.5.5 | Preconditioning and deltas for temperature cycling. |
| C2 | 1056 | 0°C to +100°C, 15 cycles, n = 22, c = 0. |
| C2 | 1051 | -55°C to +150°C, 20 cycles, n = 22, c = 0. |
| C2 | 2036 | (Not applicable to surface mount devices); lead fatigue conditions: Test condition E; .062 inch (1.57 mm) lead restriction from case. Tension conditions: Test condition A; 4 pounds, 15 seconds. |
| C5 | 4081 | $R_{\Theta JEC}$, $R_{\Theta JSP}$ = 100°C/W (maximum) +25°C ≤ $T_R$ ≤ +35°C, at $T_H$ ≥ 25 seconds in still air. (See 4.3.2.) |
| C5 | 4081 | $R_{\Theta JL}$ = 250°C/W (maximum) at .375 inch (9.52 mm) lead length, +25°C ≤ $T_R$ ≤ +35°C, at $T_H$ ≥ 25 seconds in still air. (See 4.3.2.) |
| C6 | 1038 | HTRB in accordance with 4.3.1. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein except $Z_{\Theta JX}$ need not be performed after group E subgroup 2.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.



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4.5.2 Effective carrier lifetime ($\tau_{CL}$). This test shall be measured with the device installed in the test fixture, DESC Drawing C68001 using the test set up shown on figure 5 herein. Adjust the signal generator to 54 MHz. Adjust the sensitivity of the oscilloscope to 5 mV/cm (1N5711), 10 mV/cm (1N6858), 20 mV/cm (1N5712), and 50 mV/cm (1N6857). With the device in the test fixture, adjust the output of the amplifier or VHF oscillator until the peak amplitude of the forward current is 5 cm as seen on the oscilloscope (5 mA (1N5711), 10 mA (1N6858), 20 mA (1N5712), and 50 mA (1N6857)).

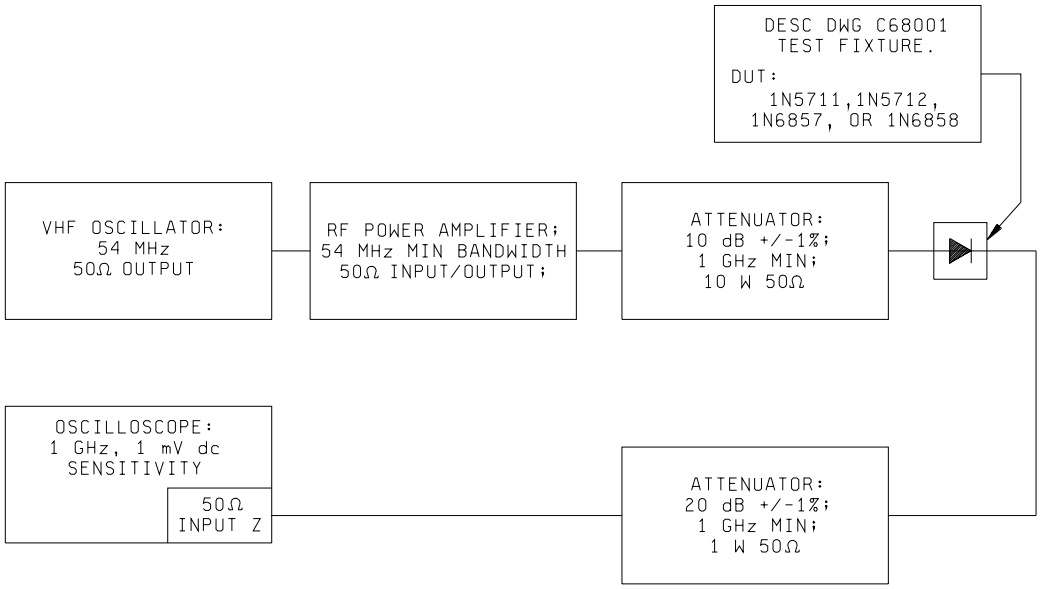



FIGURE 5. Test setup for diode effective lifetime measurement.

Change the sensitivity of scope to 2 mV/cm (1N5711, 1N6858), 5 mV/cm (1N5712), or 10 mV/cm (1N6857). Under these conditions, the effective carrier lifetime is related to the amplitude designated as "$\tau_{CL}$ " shown on figure 6. This amplitude has the calibration of 200 ps/cm (1N5711), 100 ps/cm (1N6858), 125 ps/cm (1N5712), and 100 ps/cm (1N6857).

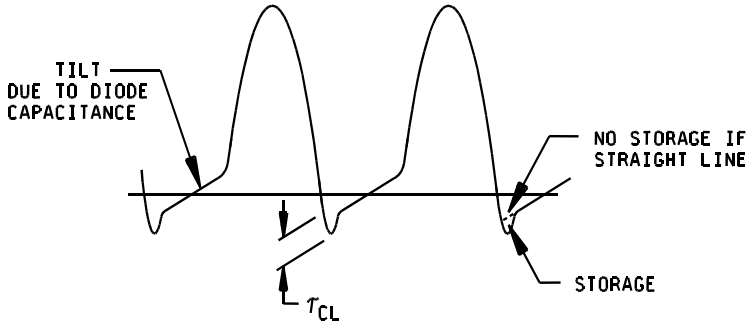



FIGURE 6. Oscilloscope display in diode effective lifetime measurement.



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4.5.3 Temperature reference point, leaded devices: LS = lead spacing = .375 inch (9.52 mm) as defined on figure 7.

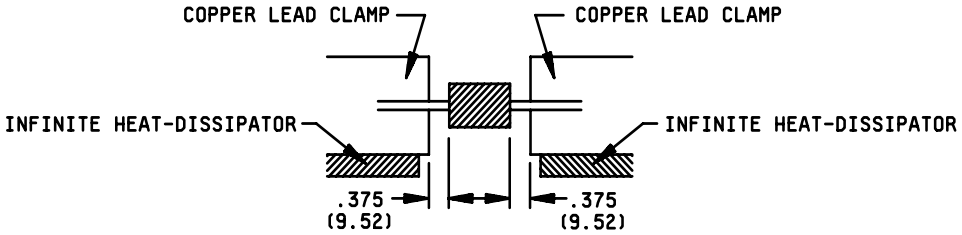



FIGURE 7. Mounting conditions.

* 4.5.4 Decap analysis (for glass packages only). Scribe and break (scribe and break not applicable to UB packages). Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification, examine the area where die (or bonding material) is in contact with the plugs, verify metallurgical bonding area.

* 4.5.4.1 Decap analysis (for UB packages only). Decap the package. Carefully remove the lid. Using 30X magnification, examine the area where die (or bonding material) is in mounted to the header, verify wire bond area.

* 4.5.5 Preconditioning and deltas for temperature cycling (not applicable to UB packages).

a. Read and record $I_{R1}$, $V_{F2}$, and $Z_{\Theta JX}$. Maintain device serialization throughout testing.

b. Immerse device terminations into any suitable flux. Flux is used to prevent solder bridging across the terminations of device and to provide for a suitable contact needed for post electrical testing.

c. Immerse device terminations into solder. The devices terminations shall be immersed instantly and not gradually, and held completely immersed in the solder for 5 +2, -0 seconds. No dwell time over the solder pot shall be permitted. Maintain solder temperature to +260°C ±5°C.

d. The device shall be allowed to return to ambient prior to cleaning. Remove residue flux from the terminations by dipping the parts in isopropyl alcohol or other suitable solvent.

e. Test devices to method 1056 and 1051 of MIL-STD-750 and in accordance with MIL-PRF-19500.

f. Read and record $I_{R1}$, $V_{F2}$, and $Z_{\Theta JX}$, test subgroup 2 of table 1 herein.

g. Failure criteria: Subgroup 2 of table 1 herein. Calculate deltas between steps 4.5.5.a and 4.5.5.f herein, $\Delta I_{R1} \leq$ 100 percent of initial reading or 25 nA, whichever is greater.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| 1N5711-1, all packages | | | | | | |
| Thermal impedance | 3101 | See 4.3.3 | $Z_{\Theta JX}$ | | 40 | °C/W |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F1}$ | | 0.410 | V dc |
| Forward voltage | 4011 | $I_F$ = 15 mA dc | $V_{F2}$ | | 1.0 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = 50 Vdc | $I_{R1}$ | | 200 | nA dc |
| Breakdown voltage | 4021 | $I_R$ = 10 μA dc | $V_{(BR)1}$ | 70 | | V dc |
| 1N5712-1, all packages | | | | | | |
| Thermal impedance | 3101 | See 4.3.3 | $Z_{\Theta JX}$ | | 40 | °C/W |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F1}$ | | 0.410 | V dc |
| Forward voltage | 4011 | $I_F$ = 35 mA dc | $V_{F2}$ | | 1.0 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = 16 Vdc | $I_{R1}$ | | 150 | nA dc |
| Breakdown voltage | 4021 | $I_R$ = 10 μA dc | $V_{(BR)1}$ | 20 | | V dc |
| 1N6857-1, UR-1 | | | | | | |
| Thermal impedance | 3101 | See 4.3.3 | $Z_{\Theta JX}$ | | 40 | °C/W |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F1}$ | | 0.350 | V dc |
| Forward voltage | 4011 | $I_F$ = 35 mA dc | $V_{F2}$ | | 0.75 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = 16 Vdc | $I_{R1}$ | | 150 | nA dc |
| Breakdown voltage | 4021 | $I_R$ = 10 μA dc | $V_{(BR)1}$ | 20 | | V dc |
| 1N6858-1, UR-1 | | | | | | |
| Thermal impedance | 3101 | See 4.3.3 | $Z_{\Theta JX}$ | | 40 | °C/W |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F1}$ | | 0.360 | V dc |
| Forward voltage | 4011 | $I_F$ = 15 mA dc | $V_{F2}$ | | 0.65 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = 50 Vdc | $I_{R1}$ | | 200 | nA dc |
| Breakdown voltage | 4021 | $I_R$ = 10 μA dc | $V_{(BR)1}$ | 70 | | V dc |
| Subgroup 3 | | | | | | |
| 1N5711-1, all packages High temperature operation | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method; $V_R$ = 50 Vdc | $I_{R2}$ | | 200 | μA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F3}$ | | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 15 mA dc | $V_{F4}$ | | 1.0 | V dc |
| Breakdown voltage | 4021 | $I_R$ = 10 μA dc | $V_{(BR)2}$ | 70 | | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 - Continued. | | | | | | |
| 1N5712-1, all packages<br>High temperature operation | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method; $V_R$ = 16 V dc | $I_{R2}$ | | 150 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F3}$ | | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 35 mA dc | $V_{F4}$ | | 1.0 | V dc |
| Breakdown voltage | 4021 | $I_R$ = 10 µA dc | $V_{(BR)2}$ | 20 | | V dc |
| 1N6857-1, UR-1<br>High temperature operation | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method; $V_R$ = 16 V dc | $I_{R2}$ | | 300 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F3}$ | | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 35 mA dc | $V_{F4}$ | | 1.0 | V dc |
| Breakdown voltage | 4021 | $I_R$ = 10 µA dc | $V_{(BR)2}$ | 20 | | V dc |
| 1N6858-1, UR-1<br>High temperature operation | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method; $V_R$ = 50 V dc | $I_{R2}$ | | 400 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F3}$ | | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 15 mA dc | $V_{F4}$ | | 1.0 | V dc |
| Breakdown voltage | 4021 | $I_R$ = 10 µA dc | $V_{(BR)2}$ | 70 | | V dc |
| Subgroup 4 | | | | | | |
| Capacitance | 4001 | | C | | | |
| 1N5711-1, 1N5712-1, and UB and UR | | $V_r$ = 0V dc; f = 1 MHz; | | | 2.0 | pF |
| 1N6857-1, 1N6858-1, and UR | | $V_{sig}$ = 50 mV(pk) maximum | | | 4.5 | pF |
| Effective carrier lifetime (see 4.5.2) 2/ | | (See DESC drawing 68001) | $\tau_{CL}$ | | 100 | ps |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ Effective carrier lifetime needs to be performed only one time per wafer lot.



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TABLE II. Group E inspection (all quality levels) - for qualification and requalification.

| Inspection 1/ | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 22 devices<br>c = 0 |
| Thermal shock | 1056 | 100 cycles, -55°C to 100°C | |
| Temperature cycling (air to air) | 1051 | 500 cycles, -55°C to 150°C | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 ($V_{F1}$, $I_{R1}$ and thermal impedance) | |
| Subgroup 2 | | | 22 devices<br>c = 0 |
| Steady-state dc blocking life | 1038 | See 4.5.1, 1,000 hrs | |
| Electrical measurements | | See table I, subgroup 2 ($V_{F1}$ and $I_{R1}$) | |
| Subgroup 4 | | | Sample size<br>N/A |
| Thermal impedance curves | | See 4.2.1.1. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices<br>c = 0 |
| ESD testing | 1020 | | |
| Subgroup 7<br>Not applicable | | | |
| Subgroup 8 | | | 45 devices |
| Resistance to glass cracking | 1057 | Test until failure occurs or to a maximum of 25 cycles, whichever comes first. | |

1/ For sampling plan, see MIL-PRF-19500.



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5711) will be identified on the QML.

| JANC ordering information | |
|---|---|
| PIN | Manufacturer |
| | 43611 |
| 1N5711-1<br>1N5712-1<br>1N6857-1<br>1N6858-1 | JANHCA1N5711, JANKCA1N5711<br>JANHCA1N5712, JANKCA1N5712<br>JANHCA1N6857, JANKCA1N6857<br>JANHCA1N6858, JANKCA1N6858 |

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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